CSCE 567 Final Visualization Project Proposal

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Project Purpose:

The purpose of this project is to inform individuals about the relationship between race, income, and political activeness (more specifically, voting) through the visualization of datasets.

I. Introduction / Background

The ideal world says that democracy is one man, one vote. However, that phrase ignores the socio-economic realities that create disparities that can be difficult to understand. Often, it is not enough to look at a single vector of society, but multiple simultaneously as well as how they interact with each other. However, doing so by just looking over datasets becomes increasingly difficult with each additional field and data point. That is why I aim to create visualizations from datasets so that correlations and trends can be more easily observed.

Two very common topics when regarding equality and American politics are race and finances. In addition to the many conversations related to these topics, I believe there is something to say regarding their relation to determining who does and does not vote during elections. I aim to explore this topic by creating visuals based on datasets containing information on average yearly family income of racial groups, percentage of a given racial group who voted, and percentage of a given family income range who voted. These visuals will be intended to show correlatory patterns in attempt to imply some sort of relation between these three topics. importantly, voting trends (such as statistics regarding what group voted for which party) is outside the projects scope.

II. Audience

The audience for my project would be those affiliated with or dedicated to political movements, ideologies, or parties—particularly those concerned with voter's rights or with encouraging people to become more active in politics.

III. Dataset(s)

Dataset 1: "Households by Total Money Income, Race, and Hispanic Origin of Householder: 1967 to 2024"
Source: Census.gov (https://www.census.gov/data/tables/2025/demo/income-poverty/p60-286.html)
URL: https://www2.census.gov/programs-surveys/demo/tables/p60/286/tableA2.xlsx
Attributes: Race, year of a given data subset (from 1967 to 2024), number of households (by the thousands) percent distribution of household income (using income ranges), median income (in dollars), and mean income (in dollars)
Number of records: (number of values for percent distribution) 4,230

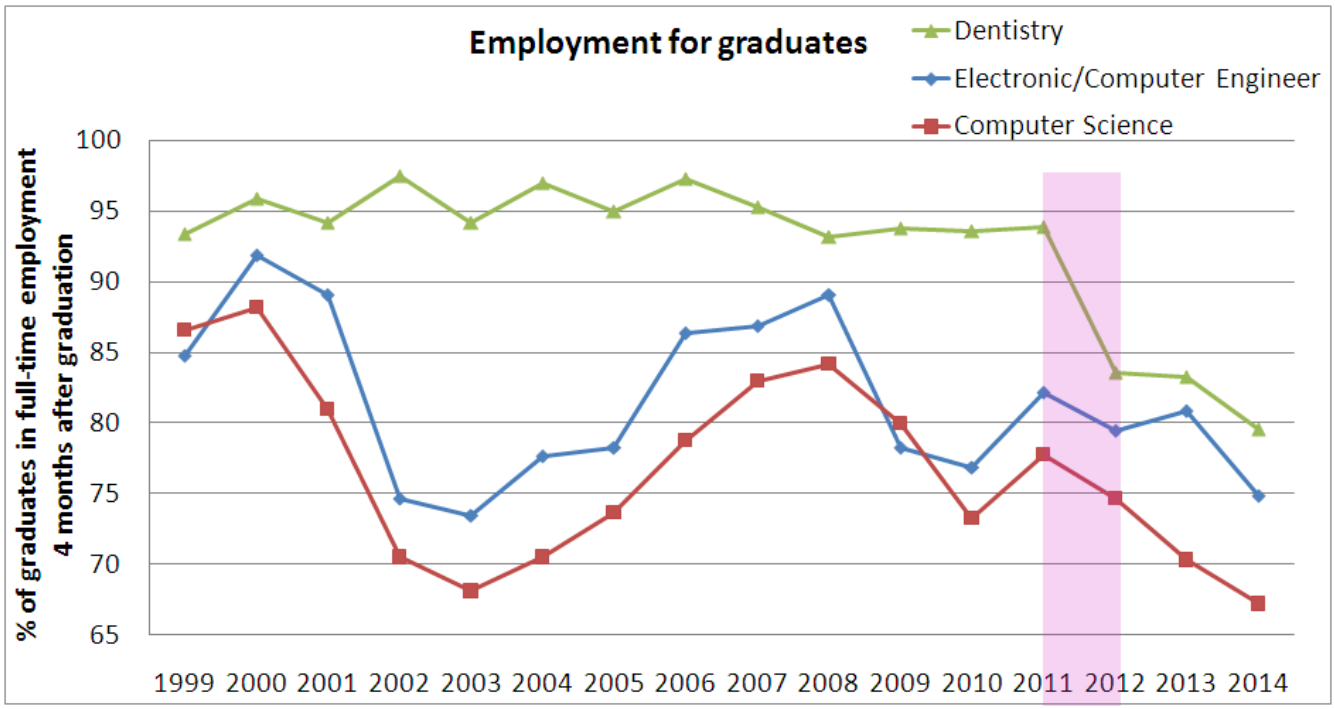
Anticipated data quality issues: One issue that I may encounter is the ranges for the household incomes may not be the same in other datasets, meaning I would have to make that clear in the key or some other part of the visualization

Dataset 2: "Reported Voting and Registration of Family Members, by Age and Family Income: November 2024"
Source: Census.gov (https://www.census.gov/data/tables/time-series/demo/voting-and-registration/p20-587.html)
URL: https://www2.census.gov/programs-surveys/cps/tables/p20/587/vote07_2024.xlsx
Attributes: age ranges, household income ranges, reported registered/not registered, reported voted/not voted, as well as an additional table showing margins of error
Number of records: (number of responses for registered/not registered and voted/not voted) 385
Anticipated data quality issues: some of the income ranges are not the same as in the previous dataset, so I will have to make note of that in the visualization. This dataset only shows responses for the 2024 election, so if I were to use it in conjunction with other datasets, I would have to only use the year 2024

Dataset 3: "Reported Voting and Registration by Race, Hispanic Origin, Sex and Age Groups: November 1964 to 2024"
Source: Census.gov (https://www.census.gov/data/tables/time-series/demo/voting-and-registration/voting-historical-time-series.html)
URL: https://www2.census.gov/programs-surveys/cps/tables/time-series/voting-historical-time-series/hst_vote01.xlsx
Attributes: year, age ranges, percent and total (in thousands) voted and registered, race. Of note, each race has a "total population" and "citizen population" sub-section. These take the percentage of the total population of a given race who voted/is registered, and the percentage of the population of a certain race who is a citizen and who voted/is registered.
Number of records: (number of percentage records for registered/voted) roughly 810
Anticipated data quality issues: a large amount of data points have no value (as they are unavailable), So in comparing different race percentages, I will have to ensure the years are available for the comparison

IV. Proposed Visualizations

Visualization group one:
A line chart that shows the change in median (or mean) income of certain races over time along with another line chart that shows change in voted percentage citizen population of those same races. The different races would be color-coded, and each chart would have keys
The interactive elements could be: if mouse-over a line, text showing the race it is representing is displayed. Additionally, it can show the value for the year the mouse is closest to (may be more difficult)

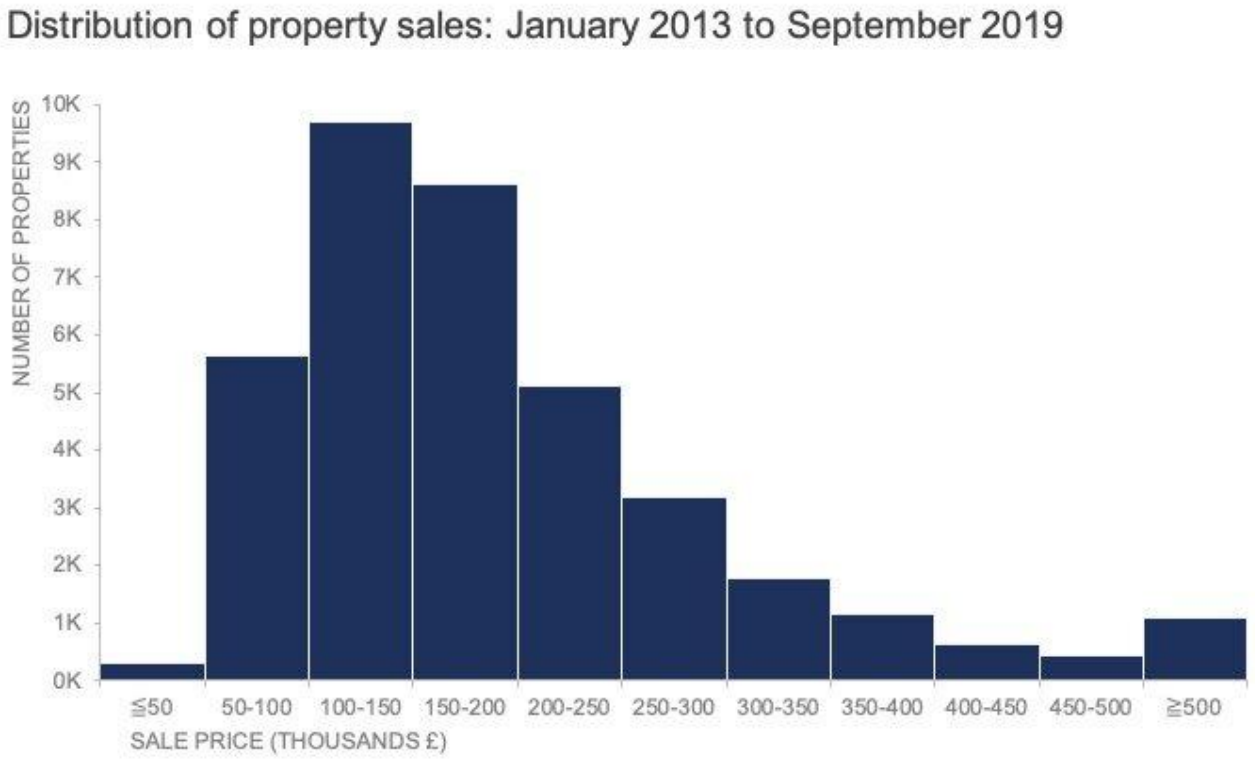


https://superuser.com/questions/878424/create-excel-chart-showing-change-multiple-industries-over-time

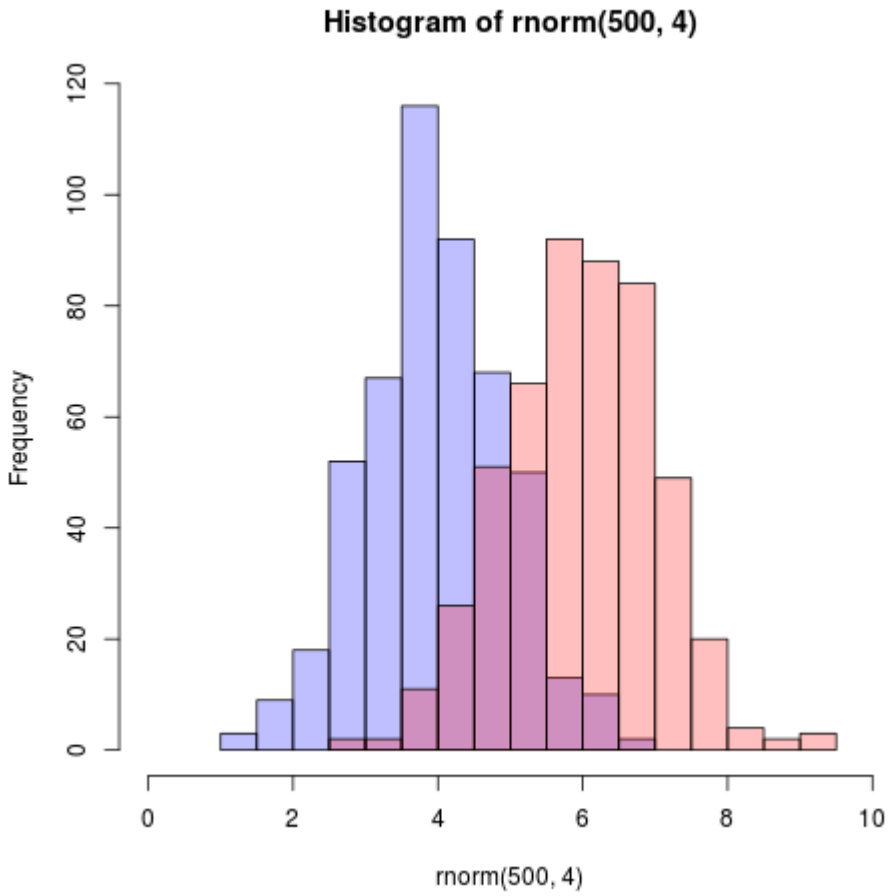
visualization group two:

An overlaid histogram with the distribution of two or more races' annual income for a given year: the different races would be color-coded and mean annual income would be shown in the key. Additionally, a standard histogram showing the distribution of the percentage of reported voters per income range for a given year. Finally, a simple bar chart that shows the percentage of the citizen population that voted for each race used in the overlaid histogram.

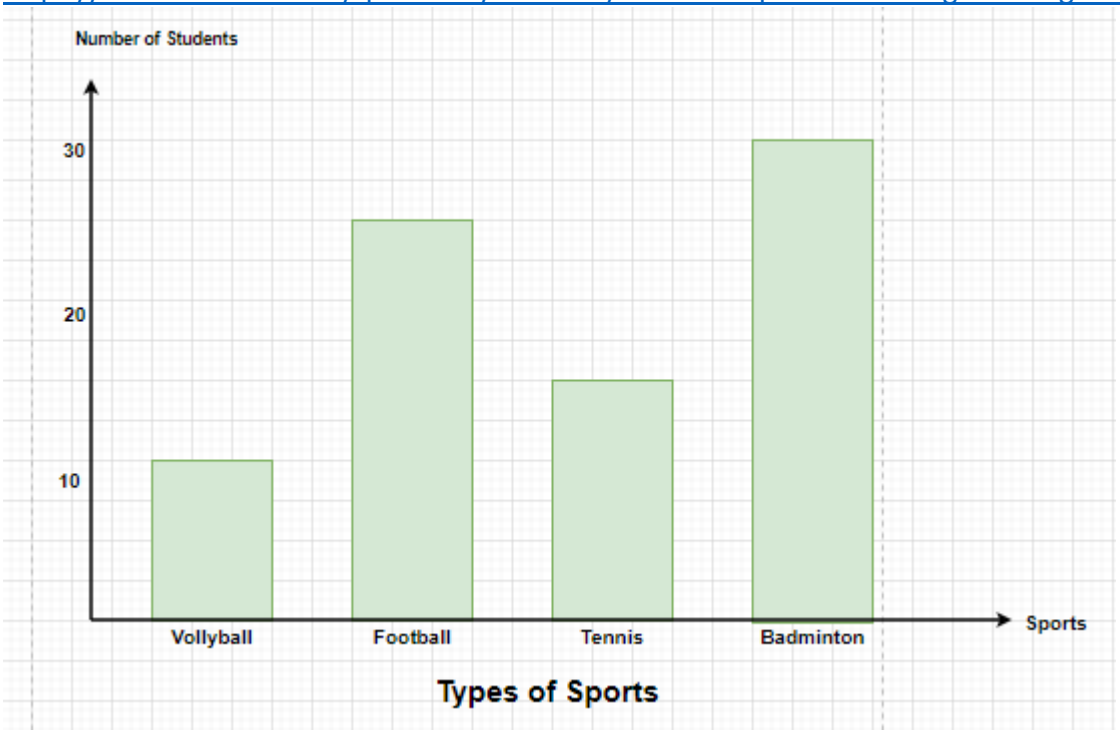
Interactive elements for the overlaid histogram would be a mini key showing the color coding for the bar that has been moused-over as well as the quantitative values for that bar. If the mouse is over two overlapping bars, the mini key will have the information from both bars. The interactive elements of the bar chart would just display the quantitative information of the mouse over-ed bar.



https://www.storytellingwithdata.com/blog/2021/1/28/histograms-and-bar-charts



https://stackoverflow.com/questions/3541713/how-can-i-plot-two-histograms-together-in-r



https://www.geeksforgeeks.org/maths/bar-graphs-and-histograms/

V. Plan

Plan A: I plan on using Python (or R if I find some visuals to be more easily made in it) and JavaScript/HTML to complete this project. I would like to deliver the visualizations through a simple webpage/HTML document that contains the visuals created with Python and/or R.

Along with the visualizations provided here, I wish to include other stand-alone visualizations that focus on a single dataset. If time permits, I would also like to graph information regarding percentage of groups that registered to vote and compare that with the percentage that did vote.

Plan B: I would make my visualizations with Tableau. I would still make a relatively basic webpage/HTML document to deliver my visualizations.

I would still try to add some stand-alone visuals. However, considering the point of my project is to find relationships between race, income, and voting, I would put most of my time towards making the visuals I have listed above/visuals that contain information about all three of the topics.